Thesis Progress: ENSO Prediction using Deep Learning

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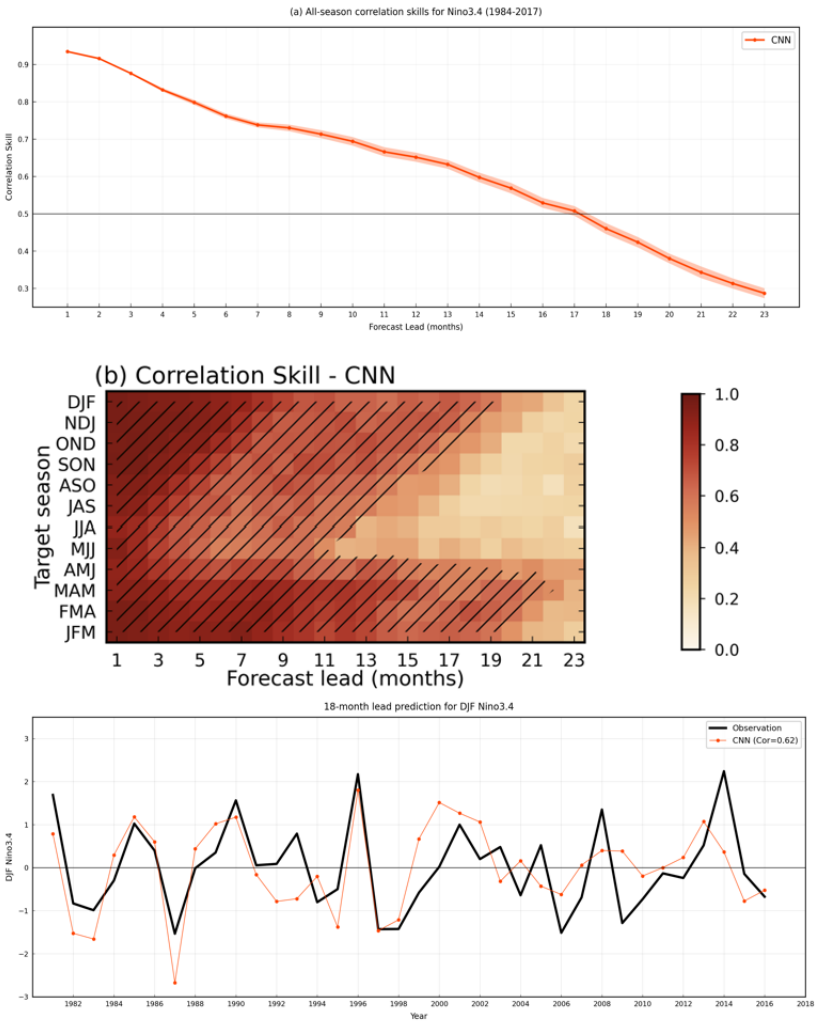
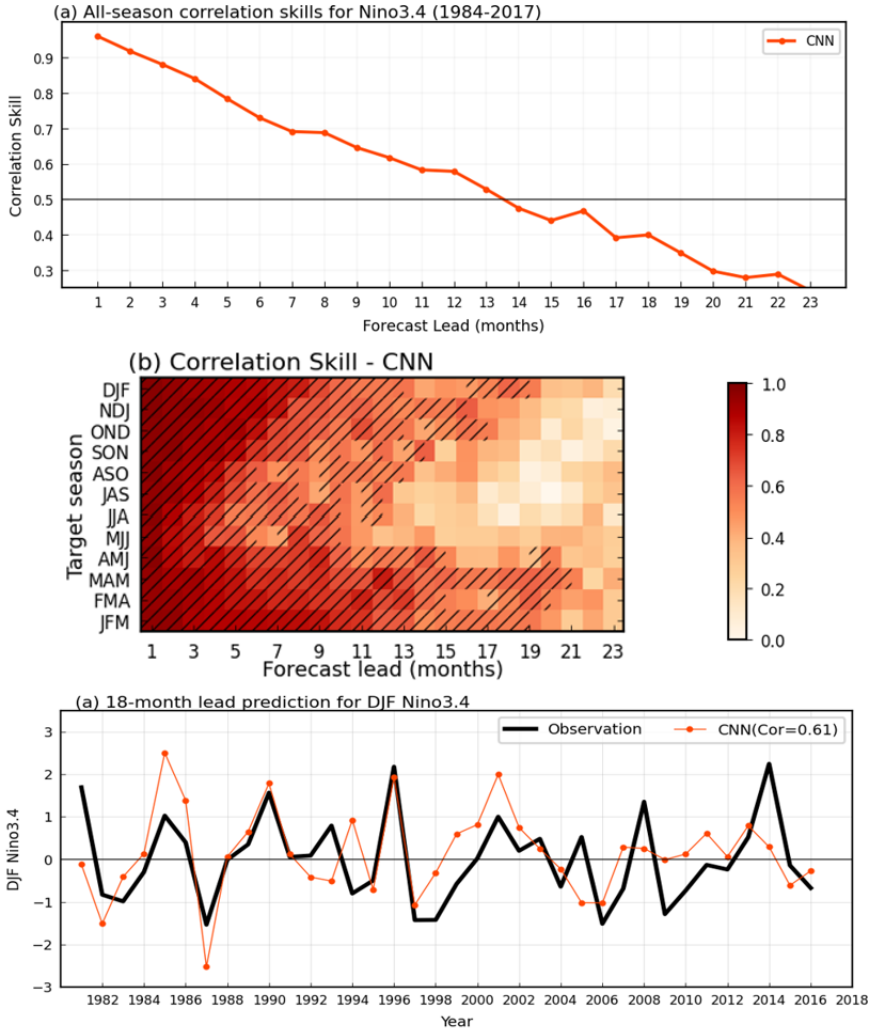
Objective

- Partial replication of the model proposed by Ham et al. (2019)
- Evaluation of:
  - Dataset: CMIP5
  - Infrastructure: CPU vs GPU
- Analysis under a consistent input setting

Comparison: Ham et al. (2019) vs IMARPE





Ham et al. (2019) original model (CPU) *CNN model trained at IMARPE (GPU)*

Experimental Setup

Scenario 1

- Dataset: CMIP5
- Input variables: SST and HC
- Infrastructure: CPU
- Lead time: 6 months

Scenario 2

- Dataset: CMIP5
- Input variables: SST and HC
- Infrastructure: GPU
- Lead time: 23 months

Computational Cost

Experimental workload

- 12 models per lead time (monthly prediction)
- 23 lead times $\Rightarrow$ 276 models
- 4 architectures evaluated
- 10 ensembles per architecture

Total models trained:

$$276 \times 4 \times 10 = 11040 \text{ models}$$

CPU Execution

- Hardware: Intel Xeon W7-2495X (24 cores)
- $\sim$30 min per model

$\approx$ 7.2 months

GPU Execution

- Hardware: NVIDIA RTX A2000 (12GB)
- $\sim$3–5 min per model

$\approx$ 35 days

Conclusions and Perspectives

Conclusions

- Successful reproduction of key figures from Ham et al. (2019)
- Consistent comparison between reference and IMARPE-trained model
- End-to-end pipeline fully implemented and validated

Perspectives (Open Questions)

- What should be the next experimental direction?
- Should the focus remain on SST and HC variables only?
- Is it worth exploring additional predictors to improve ENSO representation?